August 30, 2026

The Editor
Scientific Reports
Nature Portfolio

**Re: Submission of Original Research Article**
**"TH-DAT: Trimester-Aware Hierarchical Domain-Attention Transformer for Antenatal Depression Prediction"**

Dear Editor,

We are pleased to submit our manuscript entitled "TH-DAT: Trimester-Aware Hierarchical Domain-Attention Transformer for Antenatal Depression Prediction" for consideration as an Original Research Article in Scientific Reports.

Antenatal depression affects 10-20% of pregnant women globally and remains critically underdiagnosed, particularly in low- and middle-income countries. Current machine learning approaches for depression prediction treat clinical features as flat inputs, ignoring the clinically meaningful domain structure (demographic, obstetric, psychosocial) and the temporal evolution of risk factors across pregnancy trimesters.

Our manuscript introduces TH-DAT, a novel transformer architecture that addresses these limitations through four key innovations: (1) domain-grouped feature tokenization, (2) trimester cross-attention conditioning, (3) gated domain fusion with entropy regularization providing per-patient interpretability, and (4) two-phase training combining self-supervised pretraining with focal-contrastive fine-tuning.

TH-DAT was evaluated against 11 baseline models across four architectural categories (classical ML, deep learning, tabular transformers, and text transformers) on two geographically and culturally diverse international datasets: PERI_DEP (N=14,008, Pakistan) and Uganda EPDS (N=14,325). TH-DAT achieved the highest AUC-ROC among the evaluated models (0.9440 on PERI_DEP, 0.9026 on Uganda EPDS), with broadly comparable discrimination to Random Forest. Cross-cohort external validation, ablation analysis, multi-seed robustness testing, and calibration analysis support the reliability of the findings.

We believe this work is well suited to the scope of Scientific Reports because it presents a reproducible, methodologically sound contribution at the intersection of deep learning and maternal mental health, with potential implications for clinical screening in resource-constrained settings. The manuscript addresses a significant global health challenge with a novel, interpretable architecture validated across diverse populations.

All source code, datasets, and detailed reproducibility information are publicly available. Both datasets are openly accessible through Zenodo and Mendeley Data. The study used only pre-existing, de-identified, publicly available data and required no additional ethics approval.

This manuscript has not been published previously and is not under consideration by any other journal. All authors have read and approved the submitted version.

We confirm that this submission complies with the policies of Scientific Reports as outlined in the Guide to Authors, including the requirements for data availability, code availability, ethics, and competing interests declarations.

Thank you for considering our manuscript. We look forward to your editorial evaluation.

Sincerely,

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